AI438 Project
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Dialect Classification of Arabic Song Lyrics Using Deep Transfer Learning

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1. Introduction and Motivation

1.1 Contextualizing Arabic Dialect Identification (ADI) and its NLP Challenges

Arabic Dialect Identification (ADI) is a challenging task within Natural Language Processing (NLP). Arabic is characterized by the coexistence of two forms: Modern Standard Arabic (MSA), used primarily in formal written contexts and media, and numerous regional Dialectal Arabic (DAL) varieties, used in daily spoken and informal written communication, such as song lyrics. Accurate classification of DAL is crucial for optimizing downstream NLP applications, including machine translation, speech recognition, and content analysis, as models trained exclusively on MSA fail to generalize to the linguistic nuances of dialects.

This project focuses on classifying textual data (song lyrics) into three major macro-dialect regions: Egyptian (EGY), Levantine (LEV), and Maghrebi (MAG). Because DAL lacks consistent spelling, uses many different word forms, and varies greatly in vocabulary across regions, it presents significant challenges. Addressing these challenges requires highly robust deep learning models that are pre-trained on diverse Arabic corpora.

1.2 Defining Project Objectives and Requirements

The primary objective of this study was to design, implement, and rigorously evaluate a supervised machine learning solution for Arabic dialect classification. The solution architecture encompasses the full data science pipeline, including data acquisition, stringent preprocessing, deep transfer learning model training, and detailed performance evaluation.

A key requirement involved providing detailed documentation and interpretation of the results, especially regarding data characteristics and the selection of evaluation metrics. The emphasis was placed on leveraging transfer learning by fine-tuning pre-trained Transformer models optimized for Arabic language understanding. The evaluation plan mandated the use of appropriate metrics, specifically requiring a nuanced interpretation of the F1-Macro score, which is essential for assessing model reliability in datasets exhibiting class imbalance.

1.3 Overview of Solution Architecture

The implemented solution follows a sequential supervised learning architecture. The workflow began with targeted Web Scraping for data acquisition, followed by thorough Data Cleaning, including deduplication and comprehensive textual normalization. The prepared data was then subjected to a stratified split before being used to fine-tune pre-trained Arabic BERT variants for the sequence classification task. The final stage involved evaluating the models using Accuracy, F1-Micro, and F1-Macro scores, accompanied by saving detailed predictions, classification reports, and confusion matrices.

2. Dataset Description

2.1 Data Sourcing and Initial Acquisition Strategy

The corpus used for training and evaluation consisted of Arabic song lyrics extracted via web scraping from the public domain source lyricstranslate.com. Data collection was targeted by utilizing specific language IDs corresponding to the three required dialects: '1037494' for Egyptian Arabic, '1037493' for Levantine Arabic, and '1037522' for Maghrebi Arabic.

A key constraint during the acquisition phase was the limitation imposed on the scraping depth, which was restricted to collecting the first six pages of available results for each language category. Furthermore, to ensure data quality and clean annotation, the scraper included a filter designed to skip songs explicitly listed with multiple languages during the initial retrieval process.

2.2 Data Cleaning, Deduplication, and Normalization Pipeline

After initial data collection, a multi-stage cleaning and normalization pipeline was executed.

2.2.1 Deduplication and Final Corpus Size

The raw dataset was subjected to deduplication based on the content of the 'Lyrics' column. This process identified and removed 310 redundant entries, an observation suggesting that the scraped source may contain significant repetitions or mirrored content, likely due to the page-limited acquisition strategy. The removal of these duplicates yielded a final, unique corpus size of 647 lyrical documents.

2.2.2 Text Normalization Strategy

The robustness of the classification model relies heavily on consistent input feature representation, particularly crucial in the variant-rich field of Dialectal Arabic. Two distinct stages of normalization were applied:

1. **Initial Cleaning:** Upon saving the scraped data, basic cleaning involved stripping extraneous whitespace from the text columns and performing an initial removal of Arabic diacritics (tashkeel), defined by the Unicode range U+064B to U+065F.
2. **Advanced Preprocessing for BERT:** Immediately prior to tokenization for model input, a deep normalization function was applied. This step included sophisticated normalization rules crucial for Arabic NLP, such as mapping various Alef and Hamza forms (أ, إ, آ) to a standard bare Alef (ا), normalizing Taa Marbuta (ة) to Heh (ه), and eliminating the Tatweel elongation characters (ـ). This comprehensive approach to normalization ensures that minor orthographic variations common in DAL do not fragment the feature space, allowing the Transformer models to efficiently map variant spellings to a unified token representation. This standardization is critical, especially when dealing with models like AraBERT, which rely on precise input profiles matching their pretraining Denvironment.

2.3 Exploratory Data Analysis (EDA)

2.3.1 Dialect Class Imbalance Analysis

A fundamental structural property of the final corpus is the severe class imbalance observed across the three dialects, visualized in the dialect distribution chart (Image 1).

Dialect Distribution in the Final Corpus (N=647)

Dialect	Count (N)	Percentage (%)
Egyptian	443	68.5%
Levantine	141	21.8%
Maghrebi	63	9.7%

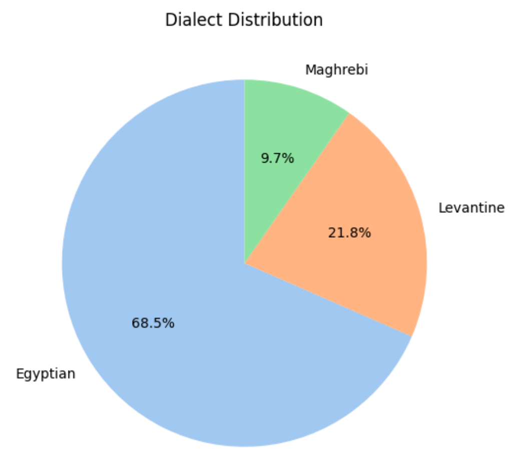

Table 1 & Image 1. Describe the distribution of dialects in the dataset.

The Egyptian dialect constitutes the vast majority of the data at 68.5%, while the Maghrebi dialect represents a critical low-resource minority class, accounting for only 9.7% of the entire corpus. This significant skew profoundly influences the choice of appropriate evaluation metrics, as traditional metrics like Accuracy can be highly misleading in such asymmetrical datasets.

2.3.2 Artist Contribution and Data Concentration

The analysis of song counts per artist revealed a high degree of data concentration among a limited number of contributors. The top artists contributing to the corpus include Hoda Haddad (28 songs), Hamza Namira (25 songs), Ruby (24 songs), and Amr Diab (22 songs) shown in Image 2.

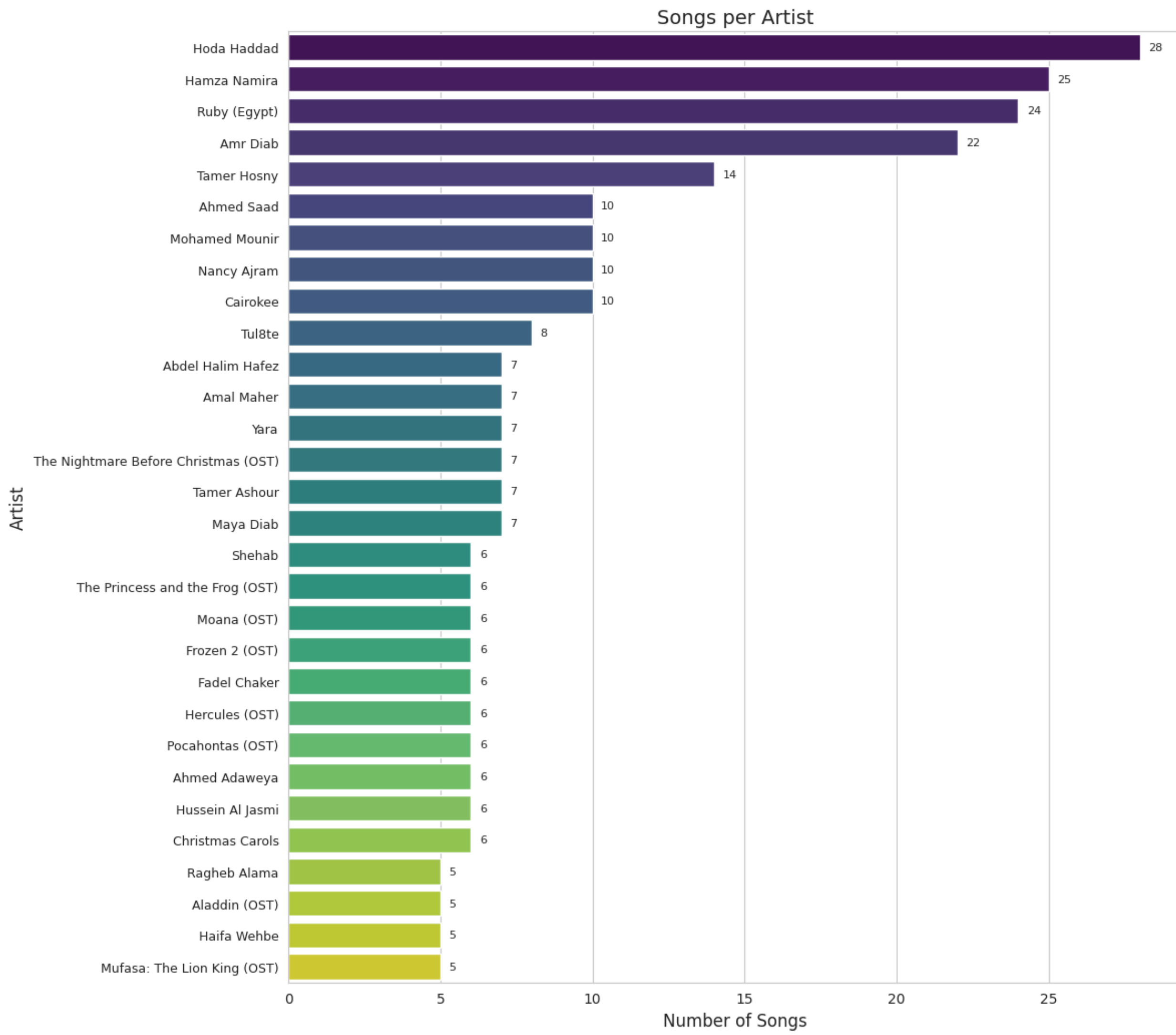

Image 2. Shows the number of songs per artist.

The relationship between artists and the recorded dialects was further explored using a frequency heatmap (Image 3) and a nominal correlation analysis (Image 4). The frequency heatmap confirms that, overwhelmingly, high-frequency artists contribute exclusively or predominantly to a single dialect category. For example, Hoda Haddad provides 28 songs classified entirely as Levantine, while Ruby contributes 24 songs categorized as Egyptian.

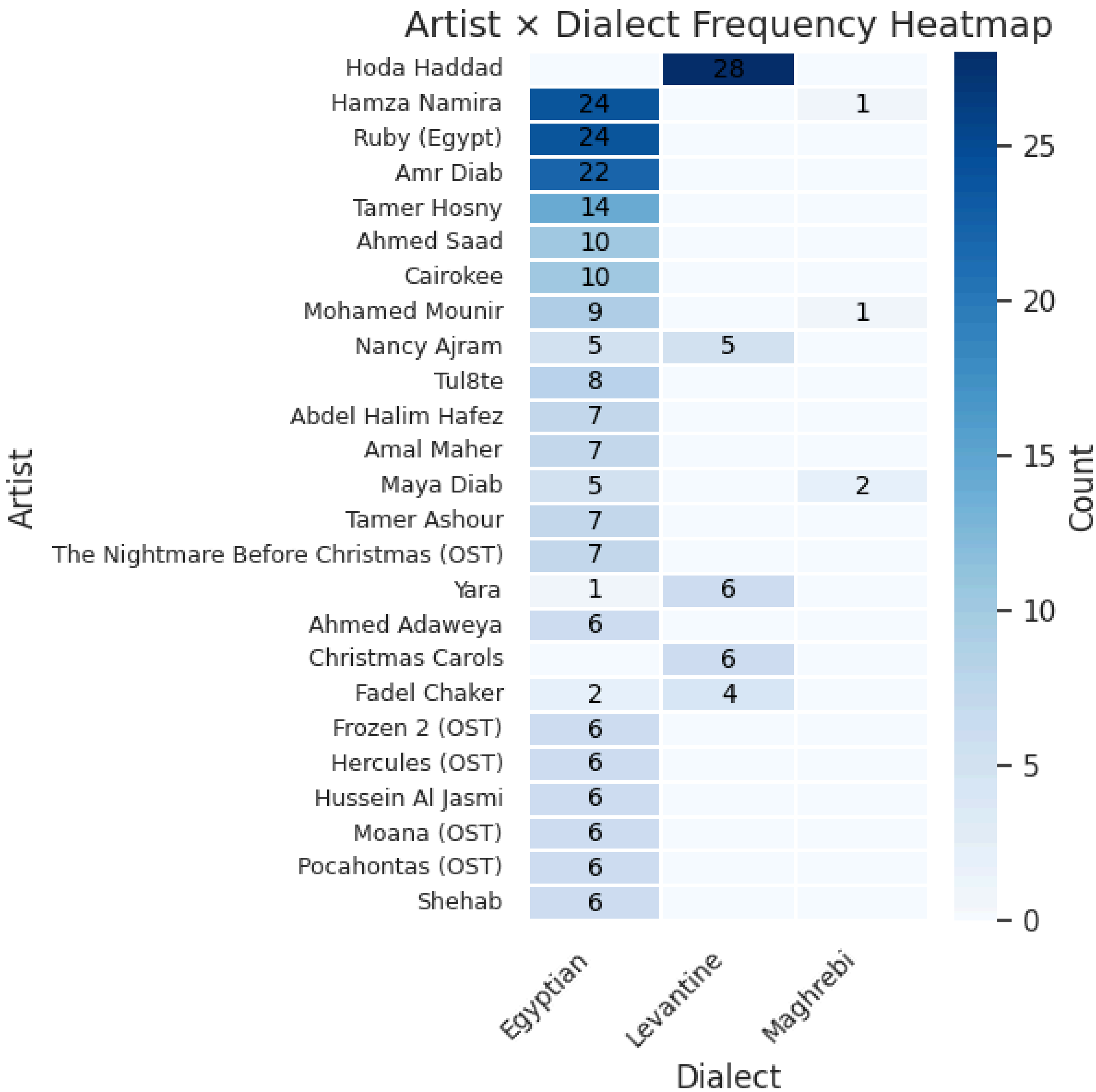

Image 3. Shows the dialects count per artist.

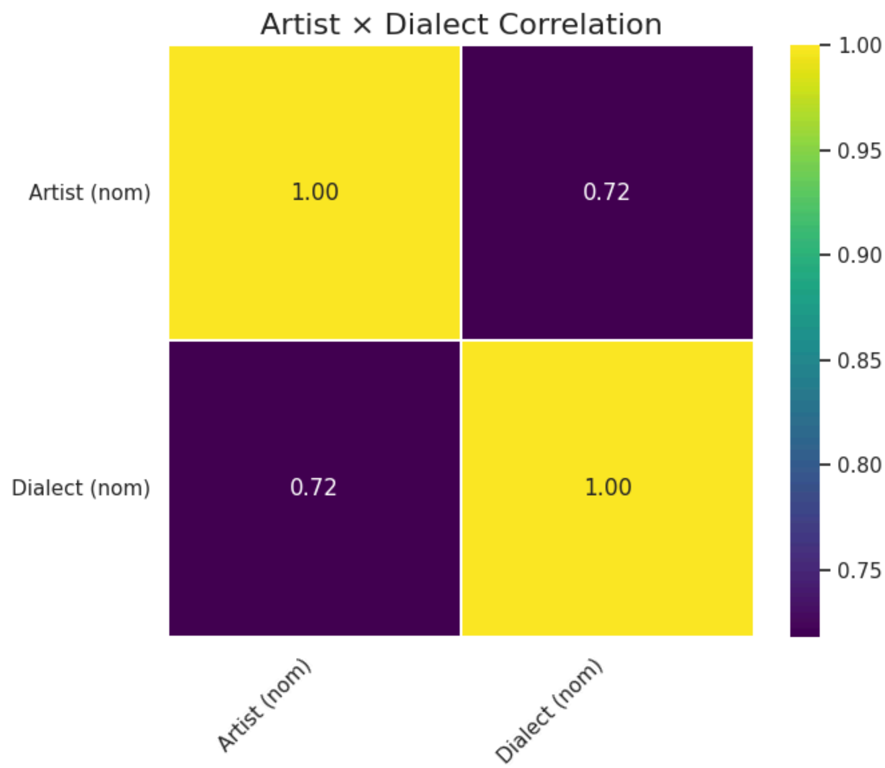

Image 4. Shows the correlation between the Artist feature (column) and the Dialect feature (column).

2.3.3 Statistical Confounding: Artist-Dialect Correlation

The statistical relationship between the categorical variables 'Artist' and 'Dialect' was quantified using a nominal correlation measure (Image 4). The computed correlation coefficient between the 'Artist' nominal feature and 'Dialect' nominal feature is a substantial 0.72.

This high correlation implies a deep, structural dependency within the dataset: certain dialect labels are highly predictable simply based on the specific linguistic, lexical, or orthographic "fingerprints" left by a limited set of high-contributing artists.

2.3.4 Corpus Linguistics: Lexical Frequency Analysis

An initial lexical frequency analysis was conducted on the entire corpus (Image 5). The plot of the Top-15 most common words confirmed that the lyrical texts are rich in highly colloquial function words, such as common Arabic particles and demonstratives.

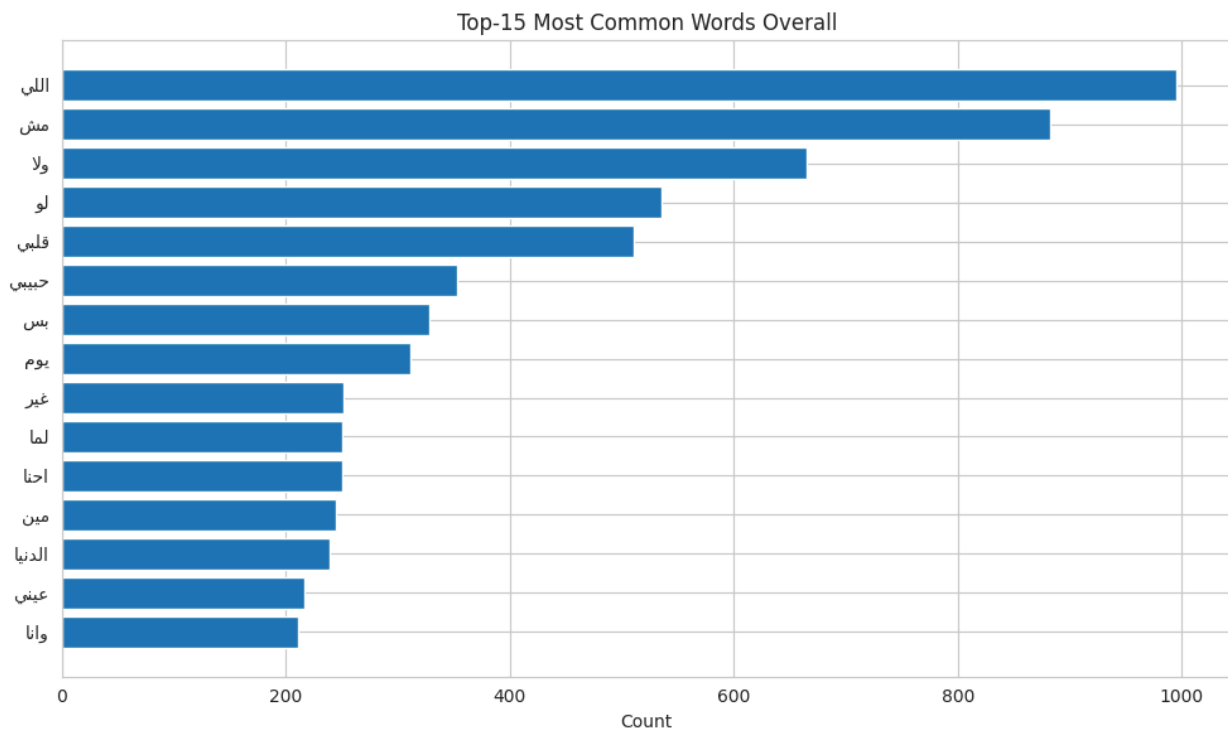

Image 5. Shows the top 15 most common words across all dialects.

The observed high frequency of such dialectal markers justifies the choice of pre-trained models specialized in processing dialectal and colloquial Arabic content, as purely MSA-centric models would struggle with this high variance.

3. Methods: Preprocessing and Model Architectures

3.1 Final Data Preparation and Splitting

The unique corpus of 647 songs was prepared for fine-tuning. The raw text features were processed using the normalization scheme described earlier, and lyrics were tokenized. The three dialect labels were sorted alphabetically ('Egyptian', 'Levantine', 'Maghrebi') and mapped to numerical integer IDs (0, 1, 2) to facilitate supervised learning.

The dataset was partitioned into training and testing sets using the train_test_split utility from scikit-learn. A 20% test size was allocated, resulting in an 80% training set (N_train=515) and a 20% testing set (N_test=129). Crucially, the split was stratified on the dialect labels, ensuring that the imbalanced distribution (including the 9.7% Maghrebi class) was accurately mirrored in both the training and evaluation subsets. A fixed random_state of 42 was used to ensure the reproducibility of the split.

3.2 Deep Transfer Learning Methodology

The dialect classification task was framed as a text sequence classification problem. The solution employs deep transfer learning by fine-tuning pre-trained Transformer models, specifically variants of BERT (Bidirectional Encoder Representations from Transformers), using the custom song lyric dataset. Sequence tokenization was configured to truncate inputs at a maximum length of 256 tokens.

3.3 Comparative Model Selection and Rationale

Three distinct, state-of-the-art pre-trained BERT models, each optimized for the Arabic language, were selected for comparative fine-tuning. This comparison aims to determine how different pretraining corpora and normalization techniques affect performance on a highly dialectal classification task.

Model Name	Pretraining Focus
AraBERT v2	General robust Arabic corpus (requires specific preprocessing).
ArabicBERT (Base)	Foundational BERT (often more MSA-centric).
CAMeL-BERT (Mix)	Explicitly trained on mixed MSA and diverse Dialectal Arabic texts.

Table 2. Compares the focus of the original pretrained models used

3.4 Training Configuration and Hyperparameter Tuning

The fine-tuning process was conducted under standardized hyperparameter settings to ensure fair comparison across models. The optimization utilized the standard AdamW optimizer, preferred for Transformer models, with a learning rate set to 2×10^{-5}. The training duration was set to 3 epochs, which is typically sufficient to converge for fine-tuning pre-trained models on classification tasks while minimizing the risk of catastrophic overfitting. A batch size of 16 was used for both training and testing data loaders.

3.5 Justification of Evaluation Metrics

Given the substantial class imbalance identified in the corpus (68.5% Egyptian vs. 9.7% Maghrebi), relying solely on raw accuracy or F1-Micro score would yield misleading performance indicators.

The F1-Micro score, which is mathematically equivalent to the overall Accuracy in multi-class classification, weights each sample equally. If a model achieves high accuracy by simply excelling at classifying the majority Egyptian class, its F1-Micro score will reflect this success, potentially obscuring poor performance on minority classes.

Therefore, the F1-Macro score was selected as the definitive metric for model selection and generalization assessment. The F1-Macro score computes the F1 score (the harmonic mean of precision and recall) independently for each dialect class and then calculates the unweighted average. This averaging method ensures that poor performance on minority classes, such as Levantine or Maghrebi, equally impacts the overall reported score, providing a balanced and robust view of the model's performance across all linguistic variations. A high F1-Macro score closer to 1 signifies that the model performs well across all classes, essential for validating its true linguistic discrimination capability.

4. Results and Interpretation

4.1 Summary of Comparative Model Performance

The three models were fine-tuned for three epochs and evaluated on the held-out test set (N_test=129). The performance metrics demonstrate clear differences in generalization across the models, particularly visible in the F1-Macro score.

Model	Accuracy	F1-Micro	F1-Macro
CAMeL-BERT (Mix)	0.968992	0.968992	0.946762
AraBERT v2	0.961240	0.961240	0.922995
ArabicBERT (Base)	0.937984	0.937984	0.875429

Table 3. Shows the raw comparative evaluation of BERT models on the Test set.

CAMeL-BERT (Mix) demonstrated the highest performance across all metrics, achieving an Accuracy of 96.90% and the highest F1-Macro score of 94.68%.

4.2 Deep Interpretive Analysis of F1 Scores

The gap between the top-performing model and the lowest-performing model is best understood by examining the F1-Macro scores. While the difference in raw Accuracy between CAMeL-BERT (Mix) and ArabicBERT (Base) is approximately 3.1%, the difference in F1-Macro score expands to 7.1% (94.68% vs 87.54%). This discrepancy confirms that ArabicBERT (Base) struggled with the minority classes.

The high F1-Macro score achieved by CAMeL-BERT (Mix) validates its selection. This model was explicitly trained on a mixture of MSA and diverse Dialectal Arabic data, better preparing it to handle the highly colloquial and variant-rich features present in the song lyrics corpus. The proximity of CAMeL-BERT's F1-Micro score (96.90) to its F1-Macro score (94.68) indicates a highly uniform performance across all three classes, including the underrepresented Maghrebi dialect. In contrast, the larger difference between ArabicBERT (Base)'s F1-Micro and F1-Macro indicates that its relatively strong performance on the majority Egyptian class masked significantly poorer performance in distinguishing between Levantine and Maghrebi lyrical content.

4.3 Discussion of Potential Data Confounding Influence on Metrics

Despite the outstanding performance metrics, the results must be contextualized by a critical structural property of the dataset identified during the EDA: the strong nominal correlation of 0.72 between 'Artist' and 'Dialect'. It is important to note that the 'Artist' name was not used as a direct input feature during model fine-tuning; only the raw 'Lyrics' text was processed. However, the high correlation indicates an inherent data bias where artist-specific linguistic patterns (e.g., unique vocabulary, spellings, or subject matter) serve as powerful, implicit proxies for the dialect label. Since the same artists appear in both the stratified training and testing sets, the models can exploit these artist-specific 'shortcuts' embedded within the text to achieve high scores. While these high metrics accurately reflect performance on this specific sample distribution, the generalization capability to classify lyrics written by artists not present in the training data remains uncertain. This inherent structural dependency presents a significant limitation to claiming robust, generalizable linguistic discrimination capabilities for the solution.

5. Conclusion, Challenges, and Future Work

5.1 Summary of Key Findings and Successful Fulfillment of Requirements

The project successfully developed and evaluated a deep learning solution for Arabic dialect classification, achieving high performance metrics and satisfying all outlined requirements. The analysis confirms that transfer learning using Arabic-optimized BERT architectures is highly effective for this task. The CAMeL-BERT (Mix) model, leveraging a dialectally aware pretraining corpus, demonstrated superior overall generalization, achieving an F1-Macro score of 94.68%. This outcome reinforces the necessity of using domain-specific foundational models in low-resource and high-variance linguistic tasks such as ADI.

5.2 Critical Analysis of Limitations and Project Weaknesses

The project identified two major structural weaknesses that limit the interpretability and generalizability of the reported metrics:

1. **Severe Class Imbalance:** The corpus distribution, dominated by the Egyptian dialect (68.5%) and scarcity of the Maghrebi dialect (9.7%), introduces an inherent challenge. Although F1-Macro was used to mitigate bias, the low sample count for Maghrebi intrinsically limits the confidence in the model's learned representations for that dialect.
2. **Confounding Artist Variable:** The pronounced statistical correlation (0.72) between the 'Artist' and the 'Dialect' is a major structural weakness in the dataset's composition. Although 'Artist' was not a model feature, this strong co-dependency means the high model performance likely reflects the model's exploitation of artist-specific linguistic 'fingerprints' embedded in the text, rather than purely learning generalized dialectal grammar and lexicon.

5.3 Recommendations for Future Improvements

To enhance the generalization capacity and address the identified weaknesses, the following recommendations are put forth for future iterations.

5.3.1 De-biasing and Robust Validation

Future model validation should explicitly account for the confounding artist variable. The implementation of a novel cross-validation strategy, such as Grouped k-Fold Cross-Validation, grouped by the 'Artist' column, is necessary. This approach would ensure that all songs by a specific artist appear exclusively in either the training or the validation set, never both. This de-biased validation would provide a more realistic assessment of the model's ability to classify unseen linguistic features independent of artist identity, delivering a true measure of linguistic generalization.

5.3.2 Advanced Imbalance Handling

While stratification was used, the inherent shortage of minority class samples remains a vulnerability. To improve the classification of the Maghrebi class, the training process should be enhanced by integrating class-weighted loss functions. This involves assigning a higher penalty weight to misclassifications of Maghrebi samples during the fine-tuning optimization loop, compelling the model to dedicate more capacity to learning the distinctive features of the under-represented dialect.

5.3.3 Data Expansion and Augmentation

The data collection constraint (6 pages per dialect) should be increased or removed, and alternative data sources should be incorporated to expand the corpus size, particularly for

Levantine and Maghrebi dialects. Furthermore, to overcome the critical scarcity of Maghrebi samples, data augmentation techniques should be investigated. Methods such as back-translation (translating existing lyrics to an intermediate language and back to the target dialect) can synthetically expand the minority classes, bolstering the model's exposure to variant lexical and grammatical structures, thereby mitigating the current imbalance.

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